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venation.

Inflorescence: The inflorescence of Euphorbiaceae varies greatly, Terminal raceme – *Croton, Ricinus,*

Catkin – *Acalypha hispida* Cyme - *Jatropha*, solitary axillary – *Phyllanthus asperulatus*, cyathium – *Euphobia* species.

Cyathium is the an unique and special inflorescence of this family. Each cyathium contains centrally a single, naked terminal female flower, usually represented by a tricarpellary gynoecium. The female flower is surrounded by a cup-like involucre formed by 4 or 5 connate sepaloid bracts. In the axil of each bract develops a group of stamens in a scorpioid manner. Each stamen represents a naked male flower because it is a jointed structure, its upper portion is the filament bearing the anther and its lower portion represents the pedicel of the male flower bearing stamen. A nectar secreting gland is present on the rim of the involucre. Glands are oval or crescent shaped and often brightly coloured. Though cyathium appears like a single flower, it actually an inflorescence.

Flowers: Flowers are always unisexual, and are highly variable. Bracteate, ebracteolate, generally unisexual, homochlamydeous, rarely heterochlamydeous, monoecious (*Baliospermum*) or dioecious (*Bridelia*), actinomorphic, rarely zygomorphic, hypogynous, rearly perigynous (*Bridelia*).

Perianth: Tepals 0 to 5 **biseriate** (male flowers of *Croton bonplandianum*) uniseriate or aphyllous (*Euphorbia*), valvate or imbricate when present, apophyllous or synphyllous.

Androecium: The number of stamens vary from 1 to many. In *Euphorbia* a single stalked stamen represents a single male flower. In *Ricinus* usually 5 stamens are present, but each stamen is profusely branched. In *Jatroph*a they are arranged in two whorls each of 5 stamens. The stamens are indefinite (*Crotons*), the filaments may be free or connate. The anthers are dithecous, dehisce either by apical or by transverse or longitudinal slits.

Gynoecium: Tricarpellary, rarely bicarpellary (*Bridelia, Mercurialis*), tetra or pentacarpellary

(*Wielandia*), syncarpous, ovary superior, rarely semi-inferior, ovules one or two in each locule on axile placentation, rarely locule splits into two forming six chambers (*Phyllanthus*). Styles 3, each split into two feathery stigma. Nectaries are usually present, gynoecium is present as a pistillode in staminate flowers.

Fruit: Fruits are capsule or schizocarp. It breaks violently and dehisce into three one seeded cocci called regma (*Ricinus*), drupe in *Emblica officinalis* and berry or samara.

Seed: Seeds are endospermous. In *Ricinus* knob-like caruncle develops from the micorpyle, that absorbs and temporarily retains water enabling germination.



Ricinus